Airand — KPI Definitions

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Summary

Twenty-three KPIs, each defined once, each computed from the tables and events in the Data Model. They answer one question for the founder — **is Zone 1 viable, and if not, what has to change?** — and one for the system admin: **is tonight going well?**

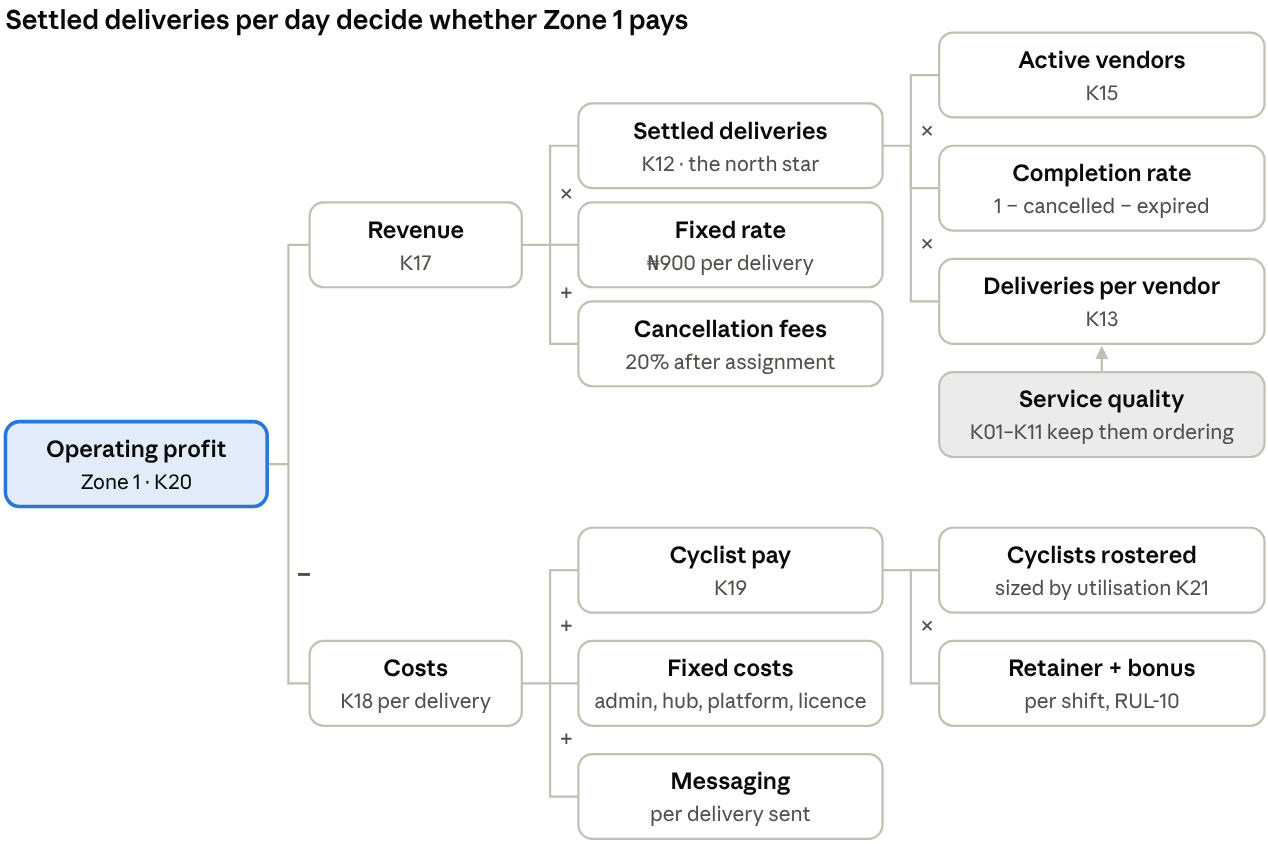
North-star measure: settled deliveries per operating day, against break-even. At the pilot's ₦900 rate and 10 cyclists, the unit-economics model puts break-even at about 46 settled deliveries a day [Assumption, Phase 2 model]. Every other KPI either moves that number (service quality keeps vendors using Airand; demand brings requests) or explains the cost side (cyclist utilisation, pay, fixed costs).

The KPIs fall into five groups:

- **Service and reliability** (K01–K11): how fast and how dependably a request becomes a delivery — the BRD's OBJ-01, OBJ-02 and OBJ-05
- **Demand** (K12–K14): how many deliveries, when, and from how many vendors — OBJ-04
- **Vendors** (K15–K16): whether vendors stay
- **Economics** (K17–K20): revenue, cost and margin per delivery and per zone — OBJ-04
- **Cyclists** (K21–K23): utilisation and reliability of the pool — BR-03

Every KPI states its formula, the events or columns it reads, its grain, its target and the requirement it serves, so the SQL in the next step has nothing to interpret.

KPI driver tree



Read right to left: each box is built from the boxes to its right.
Break-even ≈ 46 settled deliveries a day at ₦900 with 10 cyclists [Assumption].

KPI driver tree · profit back to the operational levers

The rate and the fixed costs barely move week to week, so the founder's real levers are on the right: more active vendors, more deliveries per vendor (which service quality protects), fewer cancellations, and a cyclist pool sized to demand rather than to the original plan's ten.

Counting rules

These rules apply to every KPI unless its own row says otherwise. Writing them once stops two reports disagreeing about the same night.

Rule	Definition
Request	A delivery row created by REQUEST_CREATED, whatever happened next
Activated request	Reached `active` (fee held). Expired requests never count beyond K14
Settled delivery	Ended *Delivered*, or *Failed* after pickup — the cases where the fee is settled to Airand (RUL-02, RUL-05). This is the north-star unit
Delivered	Ended *Delivered*, with or without code
Time zone	All times reported in West Africa Time (UTC+1); stored in UTC
Operating day	12:00–23:30 WAT on the calendar date. A request belongs to the day of its `created_at`
Peak	18:00–23:30 WAT, matching the roster rule that 60% of cyclist-hours fall after 18:00 (RUL-09)
Durations	Measured between event `server_ts` values; offline actions keep `device_ts` for display only
Averages	Medians and 90th percentiles for every duration — a mean hides the bad night
Trial deliveries	Count in all volume and service KPIs; contribute ₦0 to revenue (RUL-06)
Raised by the admin	Phone-in requests (FR-12) count like any other; they can be filtered out
Vendor delay	Time after the 5-minute grace at the vendor is attributed to the vendor, never to the cyclist (FR-23)
Zone	Every KPI is computed per zone, then summed or medianed for "all zones"
Data label	Any figure computed on the portfolio dataset is labelled [Simulated]

Service and reliability (K01–K11)

ID	KPI	Question it answers	Formula	Grain · cuts	Target	Serves
K01	Request-to-assignment time	How fast does a paid request get a cyclist?	CYCLIST_ACCEPTED.ts − PAYMENT_HELD.ts; median, p90 and % ⩽ 2 min	Delivery · hour, peak/off-peak	⩾ 95% within 2 min at peak	OBJ-01, BR-09
K02	Time to vendor	How long until the cyclist is at the pickup?	ARRIVED_AT_VENDOR.ts − CYCLIST_ACCEPTED.ts; median, p90	Delivery · cyclist, vendor	90% within 8 min [Assumption]	OBJ-01
K03	Vendor handover time	How long does the vendor keep the cyclist waiting?	PICKED_UP.ts − ARRIVED_AT_VENDOR.ts; median, p90	Delivery · vendor	Median ⩽ 3 min	FR-23
K04	Delivery time	How long from pickup to the door?	DELIVERED.ts − PICKED_UP.ts; median, p90	Delivered · hour, cyclist	Median ⩽ 20 min	OBJ-02
K05	End-to-end time	What does the vendor's customer experience overall?	DELIVERED.ts − PAYMENT_HELD.ts; median, p90	Delivered · vendor, hour	Tracked, no target	Case study
K06	On-time rate	Do we keep the 30-minute promise?	Delivered with K04 ⩽ 30 min ÷ delivered	Day · week, peak	⩾ 85%	OBJ-02
K07	Failure rate	How often does a picked-up package not reach the receiver?	FAILED ÷ settled deliveries; by `outcome_reason`	Day · reason, vendor	⩽ 3%	BR-13
K08	Cancellation rate	How often are requests abandoned, and when?	CANCELLED ÷ activated requests; split before assignment / after assignment / refused at pickup	Day · stage, reason, vendor	⩽ 5% overall	RUL-04, RUL-20
K09	Queue rate and queue time	How often is nobody free?	Activated requests with a QUEUED event ÷ activated; median QUEUED → CYCLIST_ACCEPTED	Day · hour	⩽ 10% of peak requests	BR-03, RUL-14
K10	Proof rate	Is every delivery closed with proof?	(DELIVERED with code + reviewed DELIVERED_WITHOUT_CODE + FAILED with reason) ÷ settled; plus without-code share	Week	100% proof; without-code ⩽ 3%	OBJ-05, BR-12
K11	Vendor delay rate	Which vendors slow handover?	Deliveries with VENDOR_DELAY_FLAGGED ÷ picked-up deliveries	Week · vendor	⩽ 10% per vendor	FR-23, statement note

Demand and vendors (K12–K16)

ID	KPI	Question it answers	Formula	Grain · cuts	Target	Serves
K12	**Settled deliveries per day** (north star)	Is the zone busy enough to pay for itself?	Count of settled deliveries per operating day; 7-day rolling average; gap to break-even	Day · week, vendor	⩾ 50/day average by month 3; break-even ≈ 46	OBJ-04, BR-03
K13	Deliveries per active vendor	Is growth coming from more vendors or busier ones?	Settled deliveries ÷ active vendors (K15), per day	Day · vendor	Tracked; anchors ⩾ 30/day	OBJ-04
K14	Demand curve and expiry	When do requests arrive, and how many never get paid?	Requests per 30-min slot; peak share = peak requests ÷ requests; expiry rate = PAYMENT_EXPIRED ÷ requests	Slot · day of week	Peak share tracked; expiry ⩽ 2%	RUL-09, RUL-02
K15	Active vendors and anchors	How many vendors actually use Airand?	Vendors with ⩾ 1 settled delivery in the last 7 days; anchors = 7-day average ⩾ 30 settled/day	Week	⩾ 8 active, ⩾ 2 anchors by month 3	OBJ-04
K16	Vendor retention and trial conversion	Do vendors stay after the free week?	Retention = vendors active in week w−1 and week w ÷ active in w−1; conversion = vendors with ⩾ 1 paid delivery in the 7 days after trial ÷ vendors whose trial ended	Week · cohort	Retention ⩾ 85%; conversion ⩾ 70% [Assumption]	BR-19, R-02

The anchor threshold of 30 settled deliveries a day comes from the BRD glossary. The Phase 3 simulated vendors suggest the busiest small vendors send closer to 15–20 a day, so this threshold is a candidate for revision once real data exists.

Economics and cyclists (K17–K23)

ID	KPI	Question it answers	Formula	Grain · cuts	Target	Serves
K17	Revenue	What did Airand actually earn?	Σ payment.amount_settled_ngn + Σ payment.cancel_fee_ngn, by settlement date; held-but-unsettled fees excluded	Day · month, vendor	⩾ ₦45,000/day by month 3	OBJ-04
K18	All-in cost per settled delivery	What does one delivery really cost us?	(K19 cyclist pay + daily fixed costs + messaging) ÷ settled deliveries; daily fixed = (monthly fixed costs ₦301,667 + cyclist kit ₦15,500 × pool size) × 12 ÷ 365, because the zone opens every day (≈ ₦15,000 a day for 10 cyclists, from the Phase 2 model [Assumption])	Day · month	Below the ₦900 rate	BR-03
K19	Cyclist pay	What did the pool cost?	Σ over shifts worked: retainer_ngn + bonus_per_delivery × max(0, settled deliveries by that cyclist that day − bonus_threshold)	Day · cyclist	Tracked against model	BR-04, RUL-10
K20	Zone operating profit and margin	Is the zone viable?	K17 − K19 − daily fixed costs − messaging; margin = profit ÷ K17	Day · month	⩾ 0 from month 3	OBJ-04
K21	Cyclist utilisation	Is the pool the right size?	Busy time ÷ shift time; busy = CYCLIST_ACCEPTED.ts → DELIVERED / FAILED / CANCELLED.ts for that cyclist; shift time = actual_end − actual_start	Shift · cyclist, hour	25–60% across the day; peak hours ⩽ 80%	BR-03
K22	Settled deliveries per cyclist per shift	Is each cyclist covering their cost?	Settled deliveries attributed to the cyclist ÷ shifts worked	Shift · cyclist	⩾ 4.6 (break-even per cyclist [Assumption])	BR-03, BR-04
K23	Cyclist unresponsive rate	Which cyclists miss offers?	ACK_TIMEOUT ÷ CYCLIST_PROMPTED, per cyclist	Week · cyclist	⩽ 5%	RUL-13

How the economics KPIs relate. K18 below ₦900 and K12 above break-even are two views of the same fact. K21 explains why: if utilisation sits under about 21% the pool is larger than demand needs, which is the Phase 2 model's warning about starting with ten cyclists. Break-even is recomputed each day from that day's actual cost base, (cyclist pay + daily fixed) ÷ (rate − messaging). Because the zone runs seven days a week rather than the model's 26 days a month, daily fixed costs are spread thinner and break-even lands nearer 39–45 than 46; the 46 is kept as the planning reference.

Targets and thresholds

The dashboard colours ten headline KPIs. Green is the pilot target; amber means watch it this week; red means act now. Thresholds are pilot defaults [Assumption] to be reviewed after four weeks of real data, alongside the timers (SRS OI-08).

KPI	Green	Amber	Red	Who acts
K12 Settled deliveries / day (7-day avg)	≥ 50	46–49	< 46 (below break-even)	Founder: vendor acquisition
K20 Zone operating profit (month to date)	≥ ₦0	−₦50,000 to ₦0	< −₦50,000	Founder: pool size, rate
K01 Assigned within 2 min at peak	≥ 95%	85–94%	< 85%	System admin: roster
K06 On-time rate	≥ 85%	75–84%	< 75%	System admin: routes, cyclists
K07 Failure rate	⩽ 3%	3–6%	> 6%	System admin: address capture
K08 Cancellation rate	⩽ 5%	5–10%	> 10%	System admin + founder
K09 Peak queue rate	⩽ 10%	10–20%	> 20%	Founder: add a cyclist at peak
K10 Without-code share	⩽ 3%	3–6%	> 6%	System admin: receiver messaging

KPI	Green	Amber	Red	Who acts
K21 Utilisation (whole day)	25–60%	21–25% or 60–70%	< 21% (pool too big) or > 70% (pool too small)	Founder: pool size
K15 Active vendors	⩾ 8	6–7	< 6	Founder: onboarding

Two thresholds pull against each other. A small pool lifts utilisation and profit (K21, K20) but raises queues at peak (K09, K01). The dashboard shows them side by side so the founder makes that trade deliberately rather than discovering it from vendor complaints.

Dashboard questions and next steps

The dashboard is organised around the questions its two users ask, not around the KPI list.

User	Question	KPIs on the page	Page
Founder	Is Zone 1 viable yet?	K12 vs break-even, K20, K17, K18	Viability
Founder	Is the pool the right size?	K21, K22, K09, K19	Viability
Founder	Are vendors staying and growing?	K15, K16, K13	Vendors
Founder	Which vendors need a conversation?	K11, K08 by vendor, K13	Vendors
System admin	When does demand hit, and are we staffed for it?	K14, K09, K01 by hour	Operations
System admin	Are we keeping the 30-minute promise?	K04, K06, K02, K03	Operations
System admin	What went wrong this week?	K07, K08, K10 with reasons	Operations
Both	Which cyclists need support?	K22, K23, K21 by cyclist	Cyclists

Next in the data track

- [ ] Generate the synthetic dataset (90 days, labelled [Simulated]) to the Data Model's schema
- [ ] Write the SQL for K01–K23 against it, one query per KPI, with the counting rules above
- [ ] Build the four-page dashboard and record what the simulated pilot shows